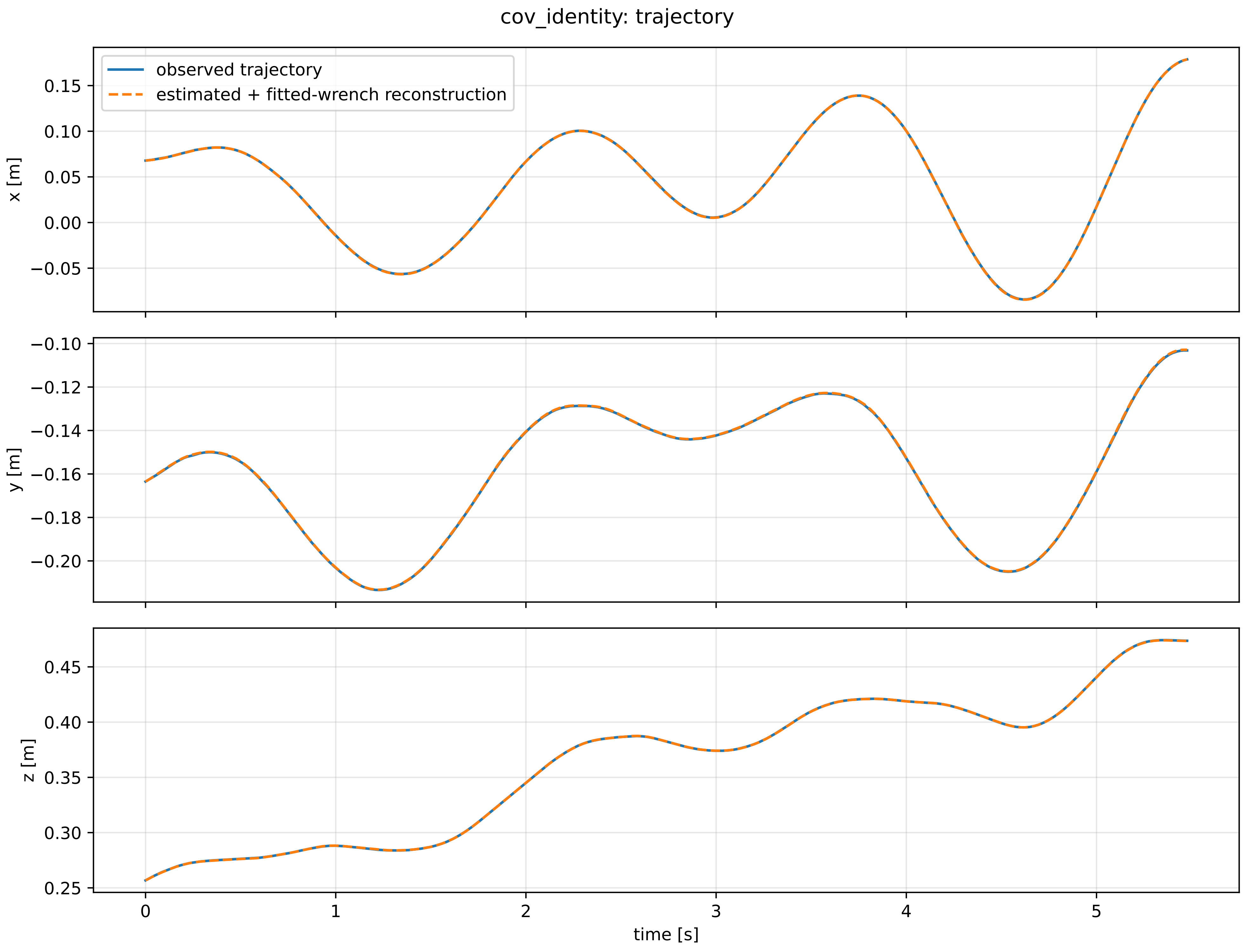
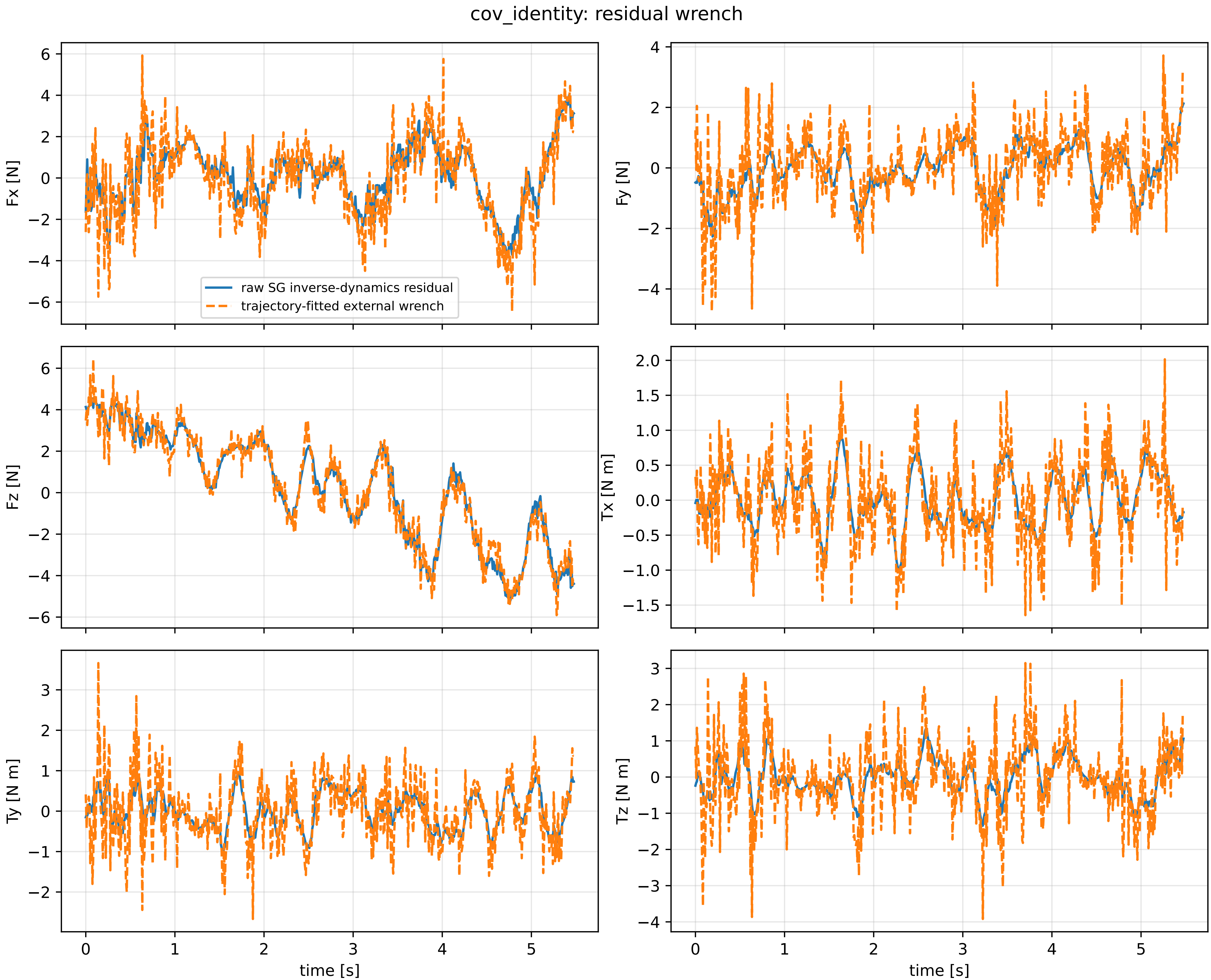
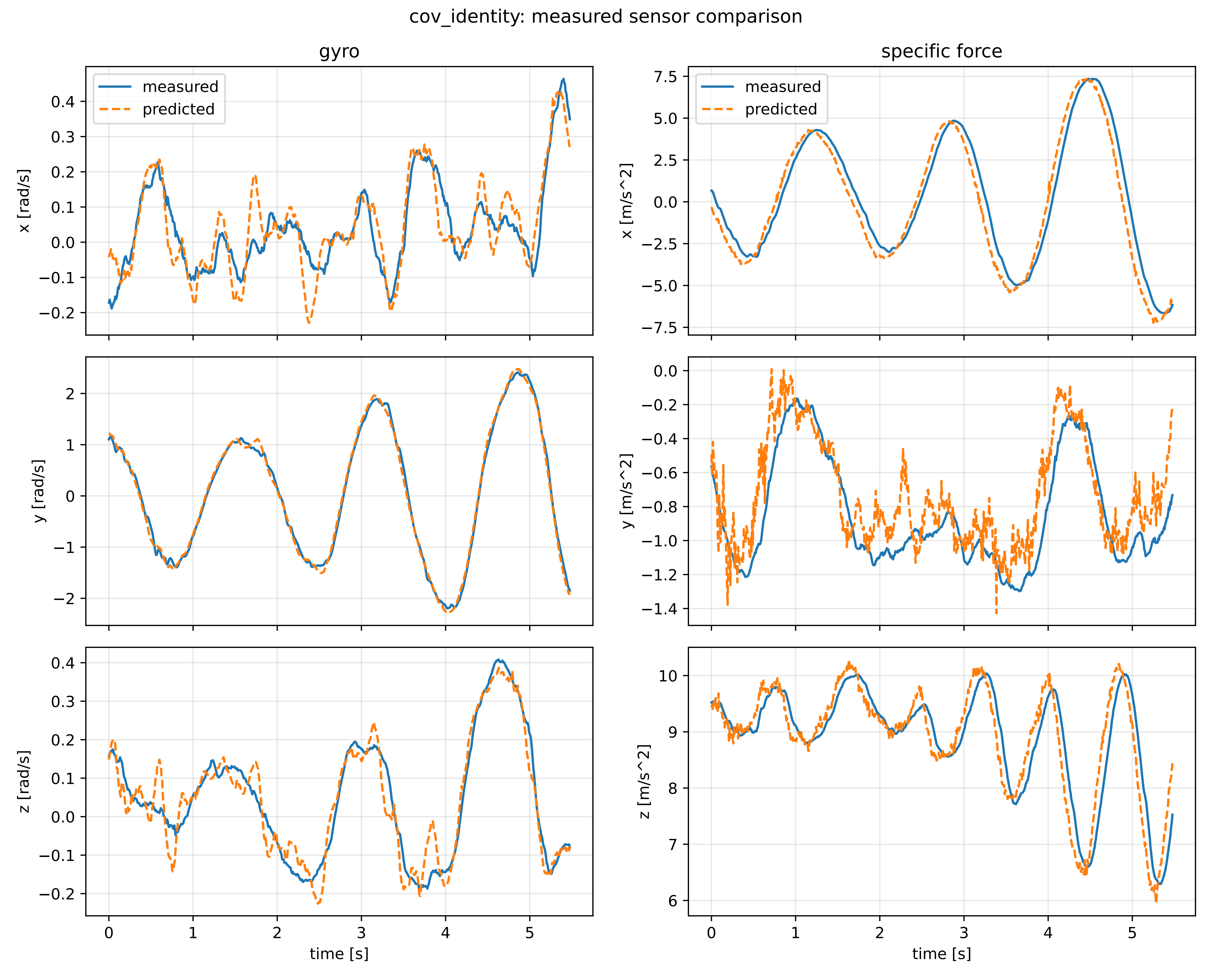


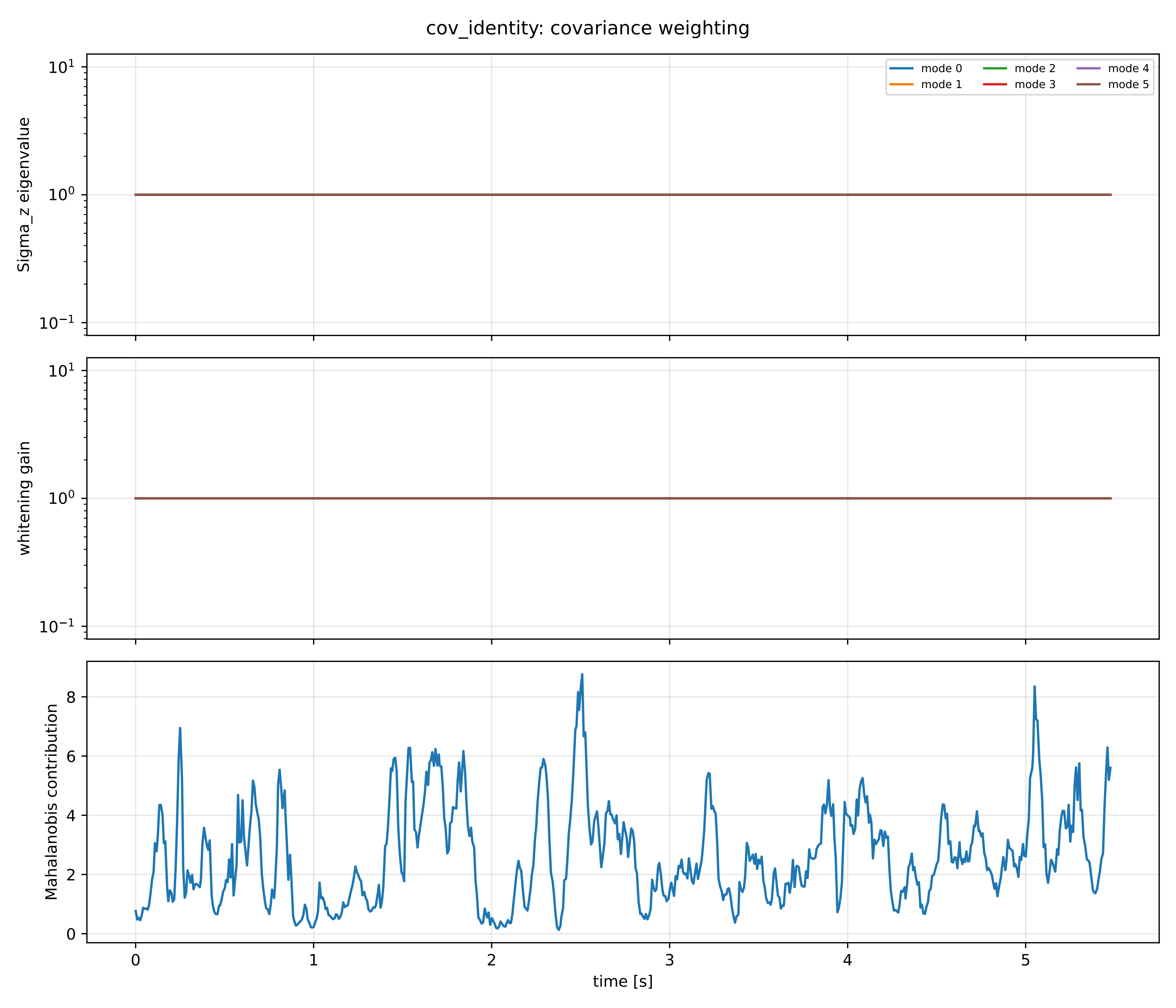
cov_identity: trajectory
observed trajectory
estimated + fitted-wrench reconstruction
x [m]
0.15
0.10
0.05
0.00
−0.05
y [m]
−0.10
−0.12
−0.14
−0.16
−0.18
−0.20
z [m]
0.45
0.40
0.35
0.30
0.25
0
1
2
3
4
5
time [s]


cov_identity: residual wrench
raw SG inverse-dynamics residual
trajectory-fitted external wrench
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time [s]
time [s]

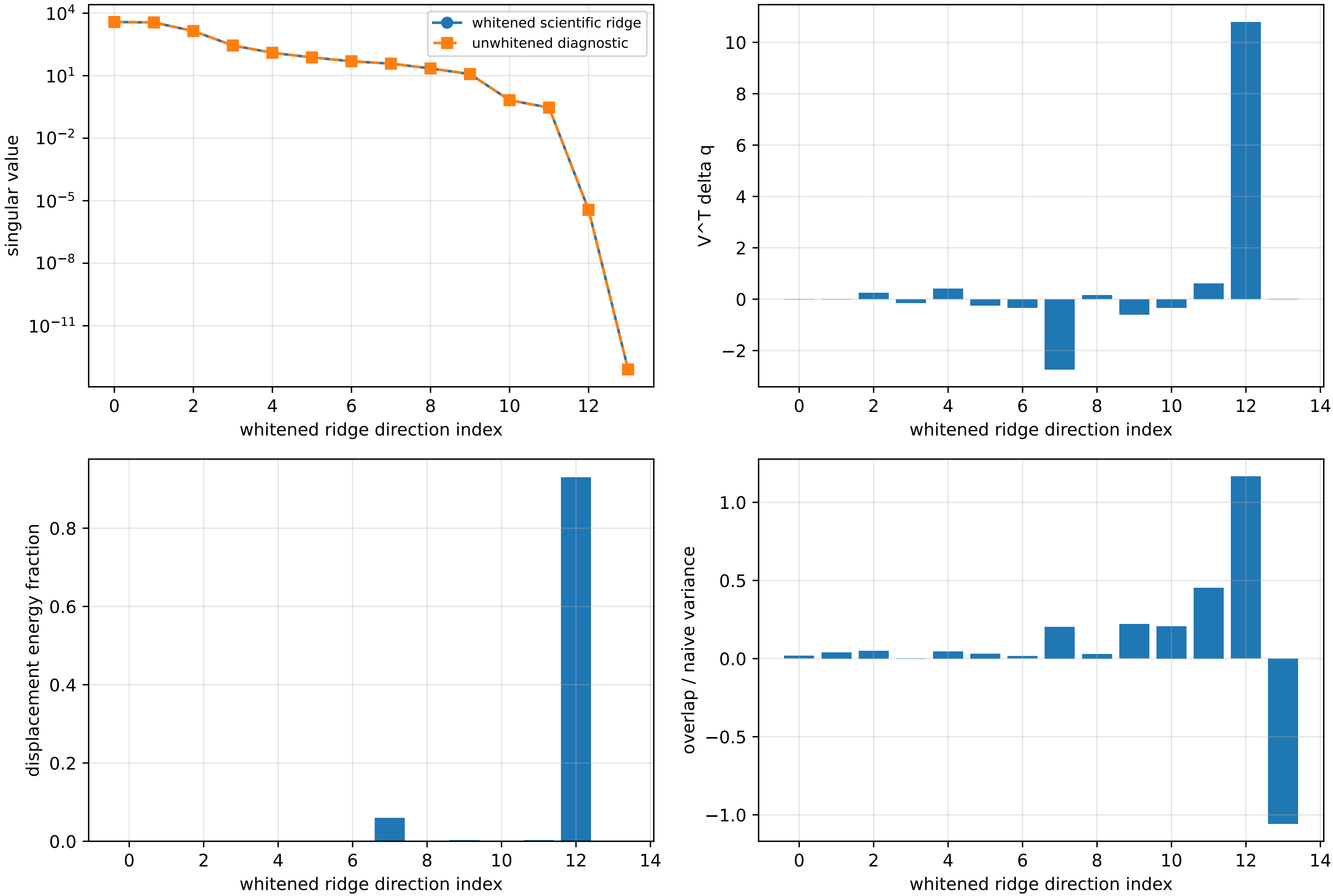

cov_identity: measured sensor comparison
gyro
specific force
measured
predicted
x [rad/s]
y [rad/s]
z [rad/s]
x [m/s^2]
y [m/s^2]
z [m/s^2]
time [s]
time [s]

cov_identity: nominal -> estimated parameters

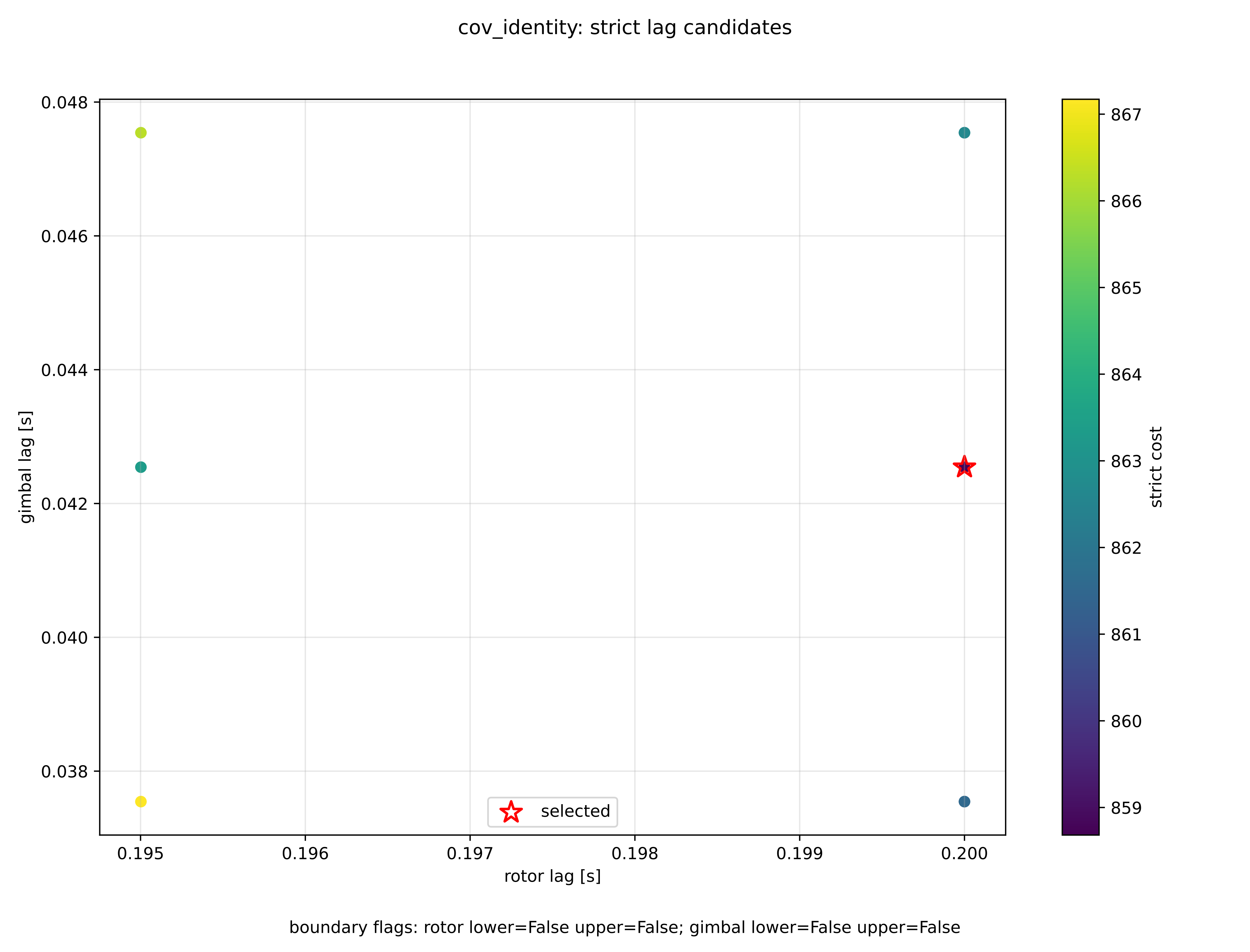
```
Case: cov_identity
status: completed (all stages completed)

mass [kg]       nominal  2.35159759   estimated  6.24729068   delta  3.89569309

inertia [kg m^2] (nominal -> estimated; delta)
  [ 6.5000061e-02 -7.2789925e-07  1.9015080e-05] -> [ 0.4184156  0.0020977 -0.0961017]   d=[ 0.3534155  0.0020984 -0.0961207]
  [-7.2789925e-07  6.4952656e-02  5.9167305e-08] -> [0.0020977 0.411558  0.0187289]   d=[0.0020984 0.3466053 0.0187288]
  [1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [-0.0961017  0.0187289  0.7805405]   d=[-0.0961207  0.0187288  0.6515484]

CoG [m]       [-2.0247086e-03 -3.0526579e-05  9.5097496e-03] -> [ 0.0083372 -0.0042761 -0.0293714]   d=[ 0.0103619 -0.0042455
rotor force effectiveness
  [1. 1. 1. 1.] -> [2.2066717 2.5449993 2.2795023 2.0310486]   d=[1.2066717 1.5449993 1.2795023 1.0310486]

rotor lag [s]  0.2
gimbal lag [s]  0.0425433703
```

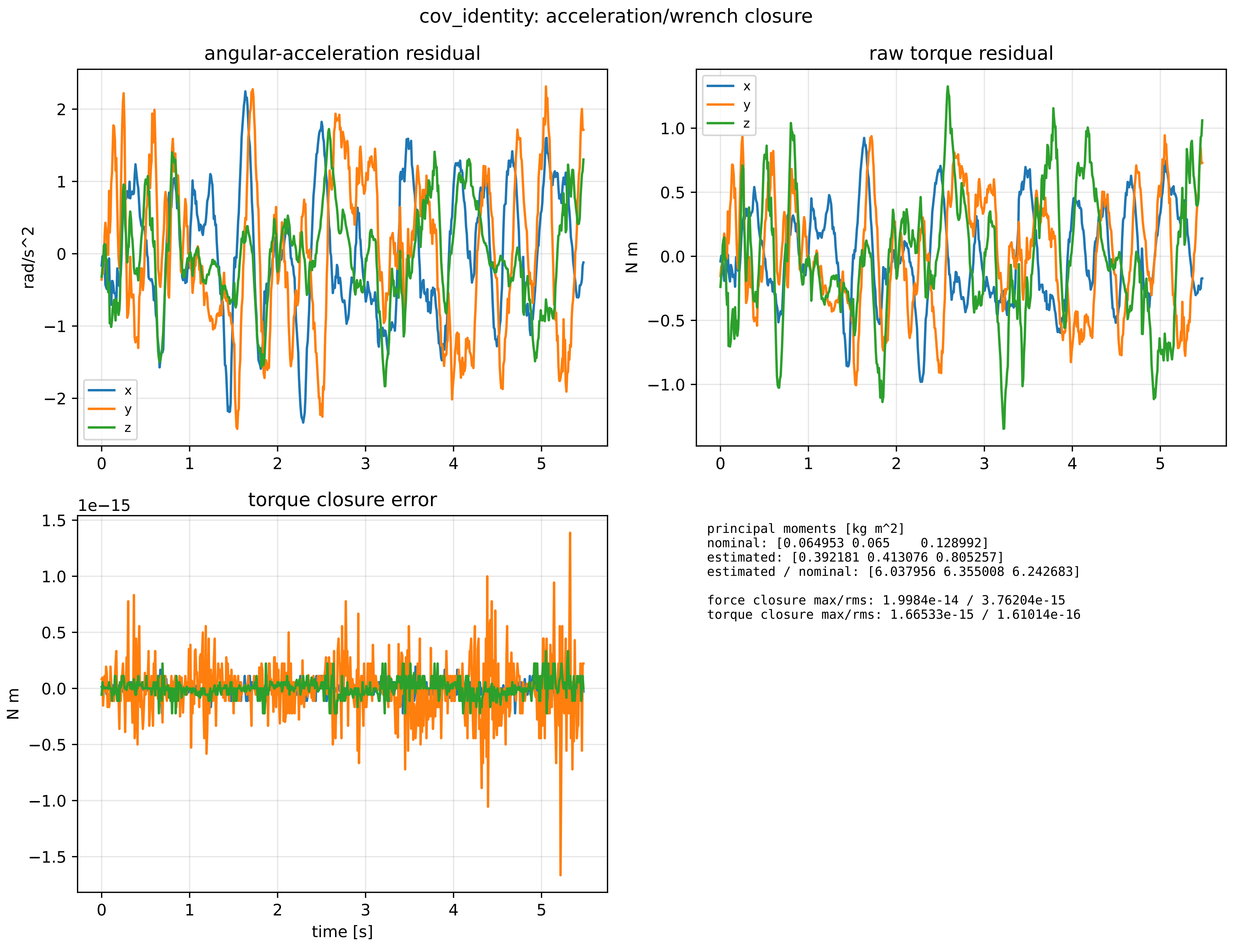

cov_identity: covariance weighting
Sigma_z eigenvalue
whitening gain
Mahalanobis contribution
time [s]
mode 0
mode 1
mode 2
mode 3
mode 4
mode 5

cov_identity: parameter ridge



exact scale gauge: ||J v_scale||=1.47611e-13, rank=13, nullity=1


cov_identity: strict lag candidates
gimbal lag [s]
rotor lag [s]
0.048
0.046
0.044
0.042
0.040
0.038
0.195
0.196
0.197
0.198
0.199
0.200
selected
strict cost
867
866
865
864
863
862
861
860
859
boundary flags: rotor lower=False upper=False; gimbal lower=False upper=False


cov_identity: acceleration/wrench closure
angular-acceleration residual
rad/s^2
raw torque residual
N m
torque closure error
1e−15
N m
time [s]
x
y
z
principal moments [kg m^2]
nominal: [0.064953 0.065   0.128992]
estimated: [0.392181 0.413076 0.805257]
estimated / nominal: [6.037956 6.355008 6.242683]
force closure max/rms: 1.9984e-14 / 3.76204e-15
torque closure max/rms: 1.66533e-15 / 1.61014e-16